

OpenClaw AI Pet Tutorial

This article defaults that the Jetson Nano Docker environment, OpenClaw expansion board, basic peripherals, and three dialogue methods are already configured. This article only expands on the AI pet application itself.

Functions in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

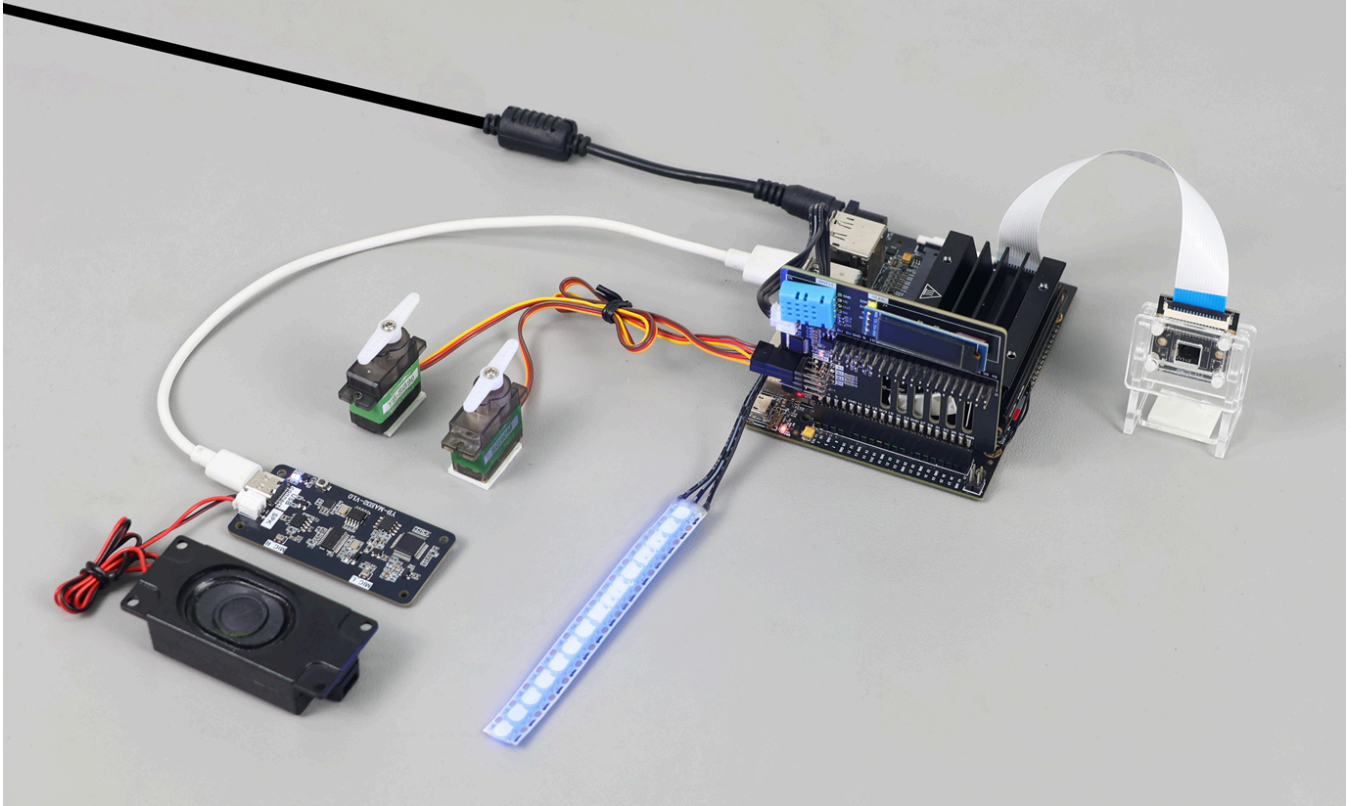
1. Tutorial Objectives

The AI pet project makes OpenClaw display more desktop pet-like interactive effects through the combination of "servo actions + RGB mood lights + OLED expressions + dialogue replies". After completion, you can achieve:

- Servo nodding, head shaking, swaying and other actions.
- RGB expresses happiness, alertness, low battery and other states.
- OLED displays expressions or short text.
- Camera observes remaining dog food situation.
- Trigger pet interaction through TUI, WhatsApp, or voice.

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- CSI camera
- RGB light strip
- Optional: Servo
- Optional: AI voice interaction module + speaker
- Wiring diagram



3. Software Entry

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Common commands:

```
/usr/local/bin/servo call-tool Servo1 --action set_angle --angle 90
/usr/local/bin/servo call-tool ServoAll --action set_all --angle1 90 --angle2 90
/usr/local/bin/rgb --format json effect --effect breathe --color magenta --speed 5
/usr/local/bin/rgb --format json color --color green
/usr/local/bin/rgb --format json color --color red
/usr/local/bin/oled text --text "Hello"
camera tool action="describe"
```

Notes:

- Servo can be used for nodding, head shaking, dancing and other actions.
- RGB can be used to express pet status.
- OLED can be used to display expressions or results.
- Camera can be used for visual gameplay such as "check how much dog food is left".

4. Basic Capability Test

Enter TUI:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔌 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)

█
```

Test first:

- Make the servo sway left and right
- Set RGB to pink breathing light
- OLED displays "Hello"

5. Three Dialogue Methods

Method	Suitable Scenario	Suggested Positioning
TUI	Local debugging actions and light effects	Preferentially verify peripheral actions execution
WhatsApp	Remote pet interaction	Suitable for remotely viewing status
Voice	Desktop pet experience	Suitable as the final demonstration method

5.1 WhatsApp Method

You can send:

- Have the servo simulate a little pet and greet me
- Take a photo to check how much dog food is left, light up green if there is, light up red if not
- Take a photo to see if there is dog food. If yes, display smile expression on OLED,make servo dance,and light up ambient lights on OLED. If no dog food,display sad expression on OLED.

20:31



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see if there is dog food. If yes, display smile expression on OLED, make servo dance, and light up ambient lights on OLED. If no dog food, display sad expression on OLED.

20:31 ✓✓



I can see the scene, but there's **no dog food** visible in the image. I can see a desk with various items like a pink mug, green fan, water bottle, and a curriculum schedule, but no dog food detected

20:31 ✓✓

Dog food check completed!

Result: No dog food detected in the scene





5.2 TUI Method

Run:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

Recommended input:

- Check if there is any dog food now



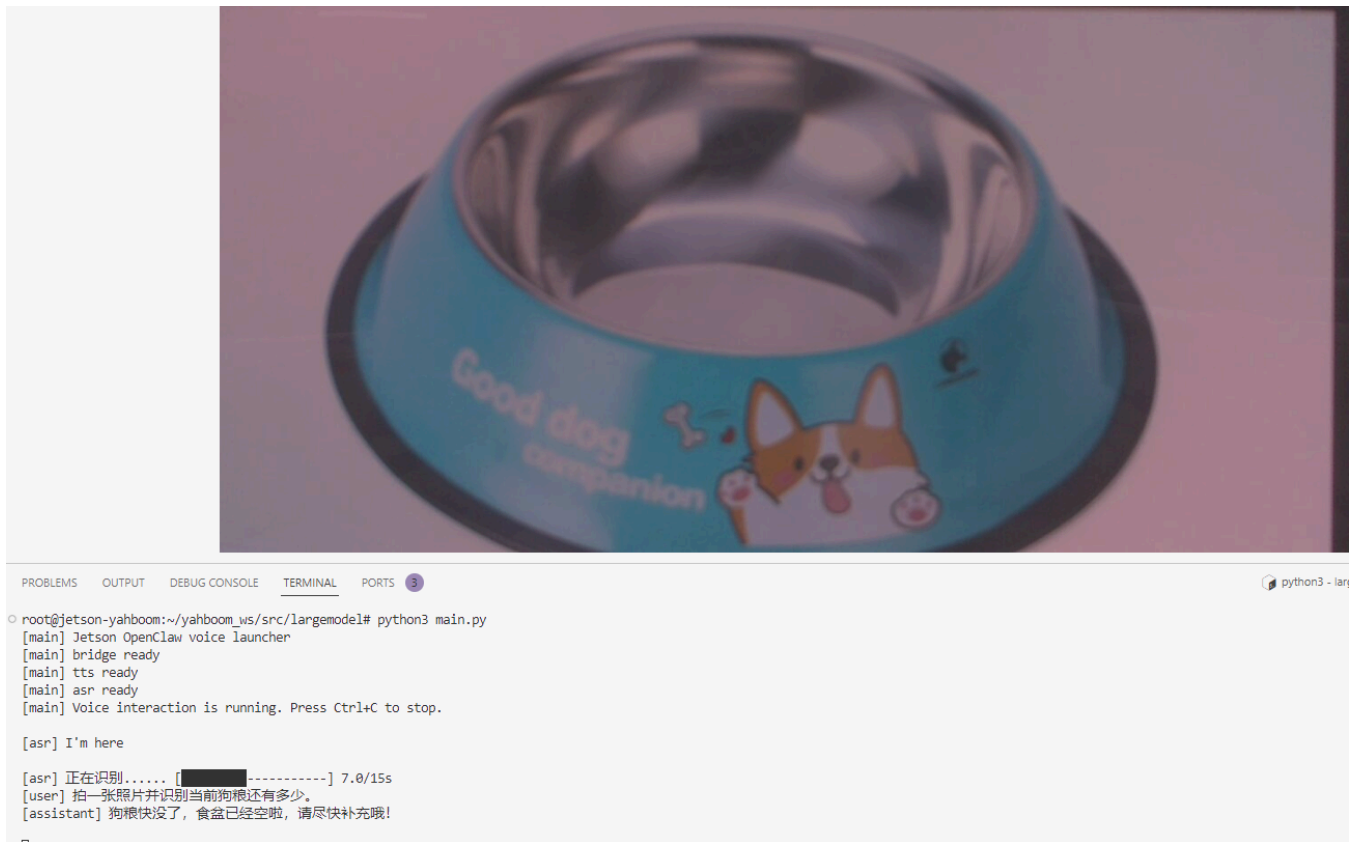
5.3 Voice Method

Execute the following command with the openclaw gateway running:

```
cd /root/yahboom_ws/src/largemodel
python3 main.py
```

You can say:

- Check how much dog food is left



6. Comprehensive Cases

6.1 Greeting

Experiment Objective: Have the AI pet use servo and OLED for welcome feedback.

Example Commands:

- Have the servo simulate a little pet and greet me
- Servo nods, OLED displays Hello
- RGB lights up pink breathing light

6.2 Emotion Expression

Experiment Objective: Express pet status with light effects and screen.

Example Commands:

- Put the pet into a happy state
- OLED displays a smiley face
- RGB shows warm-colored breathing light

6.3 Dog Food Observation

Experiment Objective: Take a photo to determine if dog food is sufficient.

Example Commands:

- Check how much dog food is left

- Take a photo to see if there's dog food
- If there's dog food light up green, if not light up red

6.4 Action Feedback

Experiment Objective: Use servo for richer action performance.

Example Commands:

- Have the servo dance for 10 seconds
- Have the servo sway left and right, with a faster rhythm
- If the answer is "yes", have the servo nod; if "no", have the servo shake its head

7. Common Problems

- Servo has no action: Check if `servo-set` or `servo-all` was generated in the reply.
- Pet has text reply but lights don't turn on: Check if RGB command was triggered.
- Dog food judgment inaccurate: Adjust camera angle and lighting to ensure dog food area is in the frame.
- Voice recognized but not executed: Check the keyword rules in `/root/yahboom_ws/src/largemodel/extension_board/openc1aw_bridge.py`.